An Exact Decomposition of the Divisor Summatory Function and Fixed-j Asymptotics

GUEHERN HWANG
Department of Mathematics, Ajou University

June 17, 2026

Abstract. In this paper, we study a decomposition of the divisor summatory function $D(x)$. We establish the exact identity

$$D(x) = 1 + \sum_{i\geq 1}(i+1)\pi(x^{1/i}) + \sum_{j\geq 2}(-1)^j(j-1)T_j(x),$$

which decomposes $D(x)$ via inclusion–exclusion according to the number of distinct prime factors of n. The three terms correspond respectively to $n = 1$, prime powers, and integers with $\omega(n) \geq 2$.

We then establish an asymptotic formula with an explicit error term for $T_j(x)$ with fixed $j \geq 2$. We consider the two-variable Dirichlet series $F(s, y)$ and obtain the factorization $F(s, y) = \zeta(s)^{2+2y}H(s, y)$, where H is holomorphic near $s = 1$. Applying the Selberg–Delange method—a truncated Perron formula, contour deformation, and local Hankel analysis—we prove that

$$T_j(x) \sim \frac{2^j}{j!}\, x\log x\,(\log\log x)^j \quad \text{as } x \to \infty.$$

These estimates describe the individual fixed-j components of the decomposition, rather than the full alternating sum.

1 Introduction

1.1 The Dirichlet divisor problem

Let $d(n)$ denote the number of divisors of a positive integer n, and let

$$D(x) := \sum_{n\leq x} d(n) \tag{1.1}$$

be its summatory function. Dirichlet [2] proved in 1849 that

$$D(x) = x\log x + (2\gamma - 1)x + \Delta(x), \tag{1.2}$$

where γ is the Euler–Mascheroni constant and $\Delta(x) = O(x^{1/2})$. The Dirichlet divisor problem asks for the infimum θ of all real numbers such that $\Delta(x) = O(x^{\theta+\varepsilon})$ for every $\varepsilon > 0$.

Hardy [3] proved that $\theta \geq 1/4$, and it is conjectured that $\theta = 1/4$. In the other direction, Voronoï [5] proved the classical estimate

$$\Delta(x) = O(x^{1/3} \log x), \tag{1.3}$$

which implies $\theta \leq 1/3$. In 2003, Huxley [4] proved the bound

$$\theta \leq 131/416. \tag{1.4}$$

In this paper, rather than pursuing further improvements of θ, we explore an alternative approach: an exact combinatorial decomposition of $D(x)$ via inclusion–exclusion on the number of distinct prime factors of n.

1.2 A new decomposition

For a positive integer n, let $\omega(n)$ denote the number of distinct prime factors of n. For each integer $j \geq 2$, define

$$T_j(x) := \sum_{n \leq x} d(n) \binom{\omega(n)}{j} \tag{1.5}$$

where $\binom{\omega(n)}{j}$ denotes the usual binomial coefficient. The main identity of this paper is the following.

Theorem 1.1. For every real number $x \geq 1$,

$$D(x) = 1 + \sum_{i \geq 1} (i+1)\pi(x^{1/i}) + \sum_{j \geq 2} (-1)^j (j-1) T_j(x) \tag{1.6}$$

where $\pi(x)$ denotes the prime-counting function. Both sums have only finitely many nonzero terms for each fixed x: we have $\pi(x^{1/i}) = 0$ for all sufficiently large i, and $T_j(x) = 0$ once j exceeds the maximum possible value of $\omega(n)$ among integers $n \leq x$.

The three terms on the right-hand side of (1.6) correspond, respectively, to $n = 1$, prime powers $\omega(n) = 1$, and integers with $\omega(n) \geq 2$. The proof of Theorem 1.1 is given in Section 2.

Dirichlet's estimate gives $D(x) \sim x \log x$. By the prime number theorem [1], the prime-power contribution satisfies

$$\sum_{i \geq 1} (i+1)\pi(x^{1/i}) \sim \frac{2x}{\log x}. \tag{1.7}$$

This is $o(x \log x)$. Thus the first two terms in the identity are negligible on the scale $x \log x$. The main contribution must therefore come from the alternating sum

$$\sum_{j \geq 2} (-1)^j (j-1) T_j(x) \tag{1.8}$$

after cancellation. As the asymptotics below show, the individual terms $T_j(x)$ are large, and substantial cancellation is needed in the full alternating sum. The fixed-j asymptotics proved below give information about individual terms, but they do not by themselves determine the full alternating sum.

Example 1.2. At $x = 10000$, direct computation gives

$$T_2(10000) = 297{,}078 \qquad T_3(10000) = 140{,}640$$
$$T_4(10000) = 26{,}674 \qquad T_5(10000) = 1{,}216$$

and $T_j(10000) = 0$ for $j \geq 6$. The prime-power contribution equals 2,711, and the identity (1.6) yields

$$1 + 2{,}711 + (T_2 - 2T_3 + 3T_4 - 4T_5) = 1 + 2{,}711 + 90{,}956 = 93{,}668 = D(10000),$$

illustrating Theorem 1.1. Moreover, the ratio of $T_j(x)$ to the leading term in Theorem 1.3 converges to 1 only very slowly: at $x = 10^9$, this ratio is 0.43 for $j = 2$ and 0.21 for $j = 3$. This slow convergence motivates the refined asymptotic expansion of $T_j(x)$ developed in Section 6.

1.3 Main results

To analyze $T_j(x)$, we consider the two-variable Dirichlet series

$$F(s, y) := \sum_{n \geq 1} \frac{d(n)(1+y)^{\omega(n)}}{n^s}, \tag{1.9}$$

which encodes $T_j(x)$ as the coefficient of y^j in the generating relation $\sum_{j \geq 0} T_j(x) y^j$. Here, we extend the definition of $T_j(x)$ to all $j \geq 0$ by the formula (1.5), so that $T_0(x) = D(x)$ and $T_1(x) = \sum_{n \leq x} d(n)\omega(n)$. The series $F(s, y)$ admits the factorization

$$F(s, y) = \zeta(s)^{2+2y} H(s, y), \tag{1.10}$$

where $H(s, y)$ is holomorphic in s on a neighborhood of $\mathrm{Re}(s) \geq 1$ and uniformly bounded for y in a small disk around 0. This factorization isolates the singularity of $F(s, y)$ at $s = 1$, and it is the starting point of the Selberg–Delange method [].

Our main asymptotic result is the following.

Theorem 1.3. For each fixed integer $j \geq 2$, as $x \to \infty$,

$$\begin{aligned} T_j(x) = {} & \frac{2^j}{j!}\, x \log x\, (\log\log x)^j \\ & + \frac{2^{j-1}}{(j-1)!}\big(\partial_y H(1,0) - 2(1-\gamma)\big) x \log x\, (\log\log x)^{j-1} \\ & + O\big(x \log x\, (\log\log x)^{j-2}\big). \end{aligned} \tag{1.11}$$

In particular,

$$T_j(x) \sim \frac{2^j}{j!}\, x \log x\, (\log\log x)^j. \tag{1.12}$$

The proof of Theorem 1.3 is given in Sections 4–6. The essential ingredients are the truncated Perron formula, contour deformation around $s = 1$, and a local Hankel analysis in the w-plane obtained via the substitution $w = (s-1)\log x$. The coefficient $\partial_y H(1, 0)$ admits an explicit expression as an absolutely convergent series over primes; we record this expression in Section 6, where it is used in the proof of Theorem 1.3.

Theorem 1.3 is a fixed-j result. Combining it with the identity (1.6) does not by itself give an asymptotic formula for $D(x)$, because the identity involves values of j that grow with x. In terms of the generating function $A(x, y) = \sum_{j \geq 0} T_j(x) y^j$, the alternating combination in the identity is governed by values near $y = -1$, whereas the analytic work below is carried out in a small disk around $y = 0$. Recovering $D(x)$ from the identity is therefore left as a future problem, requiring estimates uniform in j or a direct analysis near $y = -1$.

1.4 Outline of the paper

The paper is organized as follows. In Section 2, we prove the identity (Theorem 1.1) by partitioning $D(x)$ according to $\omega(n)$ and applying an inclusion–exclusion argument to the composite part. In Section 3, we introduce the two-variable Dirichlet series $F(s,y)$ and establish its factorization $F(s,y)=\zeta(s)^{2+2y}H(s,y)$, together with the analytic properties of $H(s,y)$ needed in the sequel.

Sections 4–6 are devoted to the proof of Theorem 1.3. In Section 4, we apply the truncated Perron formula to the generating partial sum $A(x,y):=\sum_{n\le x}d(n)(1+y)^{\omega(n)}$ and deform the contour around $s=1$. In Section 5, we carry out the local Hankel analysis via the substitution $w=(s-1)\log x$ and evaluate the main and lower-order contributions. In Section 6, we extract the asymptotic formula for $T_j(x)$ from that of $A(x,y)$ by differentiating in y, thereby completing the proof of Theorem 1.3.

Finally, in Section 7, we discuss why the fixed-j asymptotics and the identity do not yet determine the full alternating sum for $D(x)$, and what additional uniform estimates would be needed.

2 Proof of the identity

In this section, we prove the identity stated in Theorem 1.1. The proof proceeds by partitioning the sum $D(x)=\sum_{n\le x}d(n)$ according to the value of $\omega(n)$, and then applying an inclusion–exclusion argument to the composite part.

Proof of Theorem 1.1. For fixed x, all sums below are finite. Indeed, $\pi(x^{1/i})=0$ for all sufficiently large i, and $\binom{\omega(n)}{j}=0$ whenever $j>\omega(n)$.

Partition the integers n with $1\le n\le x$ into three disjoint classes:

(i) $n=1$ (equivalently, $\omega(n)=0$);

(ii) prime powers, i.e., $\omega(n)=1$;

(iii) composite integers with $\omega(n)\ge 2$.

Writing $D(x)$ as the sum of contributions from each class, we have

$$D(x)=d(1)+\sum_{\substack{n\le x\\ \omega(n)=1}}d(n)+\sum_{\substack{n\le x\\ \omega(n)\ge 2}}d(n). \tag{2.1}$$

Since $d(1)=1$, the first term equals 1.

For the second term, every integer n with $\omega(n)=1$ is of the form $n=p^i$ for a unique prime p and a unique integer $i\ge 1$, and $d(p^i)=i+1$. The number of prime powers $p^i\le x$ with exponent exactly i equals $\pi(x^{1/i})$. Hence

$$\sum_{\substack{n\le x\\ \omega(n)=1}}d(n)=\sum_{i\ge 1}(i+1)\pi(x^{1/i}). \tag{2.2}$$

For the third term, we apply inclusion–exclusion on the set of distinct prime divisors of n. For each integer $j\ge 2$, recall the definition

$$T_j(x)=\sum_{n\le x}d(n)\binom{\omega(n)}{j}.$$

We claim that

$$\sum_{\substack{n\le x\\ \omega(n)\ge 2}} d(n) = \sum_{j\ge 2}(-1)^j (j-1)T_j(x). \tag{2.3}$$

To prove (2.3), interchange the order of summation on the right-hand side. This involves only finite sums, so no convergence issue occurs:

$$\sum_{j\ge 2}(-1)^j (j-1)T_j(x) = \sum_{n\le x} d(n) \sum_{j\ge 2}(-1)^j (j-1)\binom{\omega(n)}{j}.$$

For n with $\omega(n)\in\{0,1\}$, we have $\binom{\omega(n)}{j}=0$ for all $j\ge 2$, so such n do not contribute. For n with $\omega(n)=m\ge 2$, we use the following combinatorial identity.

Lemma 2.1. For every integer $m\ge 2$,

$$\sum_{j=2}^{m}(-1)^j (j-1)\binom{m}{j} = 1. \tag{2.4}$$

Proof of the lemma. Starting from the standard identities

$$\sum_{j=0}^{m}(-1)^j\binom{m}{j} = 0 \quad \text{and} \quad \sum_{j=0}^{m}(-1)^j\, j\binom{m}{j} = \frac{d}{dx}(1-x)^m\bigg|_{x=1} = 0 \text{ (for } m\ge 2),$$

we combine them to obtain

$$\sum_{j=0}^{m}(-1)^j (j-1)\binom{m}{j} = 0 \quad (\text{for } m\ge 2).$$

Separating the $j=0$ and $j=1$ terms:

$$1\cdot(-1) + 0\cdot\binom{m}{1} + \sum_{j=2}^{m}(-1)^j (j-1)\binom{m}{j} = 0,$$

which yields

$$\sum_{j=2}^{m}(-1)^j (j-1)\binom{m}{j} = 1. \qquad \square$$

Applying the lemma, we obtain

$$\sum_{n\le x} d(n)\sum_{j\ge 2}(-1)^j (j-1)\binom{\omega(n)}{j} = \sum_{\substack{n\le x\\ \omega(n)\ge 2}} d(n)\cdot 1,$$

which establishes (2.3).

Combining the contributions from all three classes yields

$$D(x) = 1 + \sum_{i\ge 1}(i+1)\pi(x^{1/i}) + \sum_{j\ge 2}(-1)^j (j-1)T_j(x),$$

completing the proof. $\square$

3 Dirichlet series and factorization

In this section, we introduce the two-variable Dirichlet series $F(s, y)$ and establish its key analytic properties. The main output is the factorization $F(s, y) = \zeta(s)^{2+2y} H(s, y)$, where $H(s, y)$ is holomorphic in a neighborhood of $s = 1$ and uniformly bounded in a small disk around $y = 0$.

3.1 Euler product

Recall the definition

$$F(s, y) := \sum_{n \geq 1} \frac{d(n)(1+y)^{\omega(n)}}{n^s}. \tag{3.1}$$

Since the arithmetic function $n \mapsto d(n)(1+y)^{\omega(n)}$ is multiplicative in n for each fixed y, the series admits an Euler product representation for $\mathrm{Re}(s) > 1$ [1].

Lemma 3.1. For $\mathrm{Re}(s) > 1$ and every $y \in \mathbb{C}$,

$$F(s, y) = \prod_p \left(1 + (1+y)\left((1-p^{-s})^{-2} - 1\right)\right), \tag{3.2}$$

where the product is over all primes p.

Proof. By multiplicativity, for $\mathrm{Re}(s) > 1$ we have

$$F(s, y) = \sum_{n \geq 1} \frac{d(n)(1+y)^{\omega(n)}}{n^s} = \prod_p \left(\sum_{k \geq 0} \frac{d(p^k)(1+y)^{\omega(p^k)}}{p^{ks}} \right).$$

For $k = 0$, the summand equals 1. For $k \geq 1$, we have $d(p^k) = k + 1$ and $\omega(p^k) = 1$, so the local factor becomes

$$1 + (1+y) \sum_{k \geq 1} \frac{k+1}{p^{ks}}.$$

Setting $u := p^{-s}$ and using the standard identity

$$\sum_{k \geq 1} (k+1) u^k = \frac{1}{(1-u)^2} - 1 \quad \text{for } |u| < 1,$$

the local factor equals $1 + (1+y)\big((1-p^{-s})^{-2} - 1\big)$, which proves (3.2). □

3.2 Factorization isolating the singularity

The Riemann zeta function admits the Euler product [1] $\zeta(s) = \prod_p (1-p^{-s})^{-1}$ for $\mathrm{Re}(s) > 1$, so that $\zeta(s)^{2+2y} = \prod_p (1-p^{-s})^{-(2+2y)}$ on the same half-plane, provided a branch of the logarithm is fixed. We use the principal branch, which is analytic in s in a neighborhood of any point with $\mathrm{Re}(s) > 1$ and in y in a neighborhood of 0.

Define

$$H(s, y) := \prod_p (1-p^{-s})^{2+2y} \left(1 + (1+y)\left((1-p^{-s})^{-2} - 1\right)\right). \tag{3.3}$$

Theorem 3.2. For $\mathrm{Re}(s) > 1$ and y in a neighborhood of 0,

$$F(s, y) = \zeta(s)^{2+2y} H(s, y). \tag{3.4}$$

Proof. This follows by combining Lemma 3.1 with the Euler product for $\zeta(s)^{2+2y}$ and regrouping factors. $\square$

The purpose of the factorization (3.4) is to isolate the singular behavior of $F(s, y)$ at $s = 1$ into the factor $\zeta(s)^{2+2y}$, leaving $H(s, y)$ as a well-behaved function in a larger region, as we now establish.

3.3 Analytic properties of $H(s, y)$

We show that the Euler product (3.3) defines a holomorphic function on a strip containing $\mathrm{Re}(s) = 1$, uniformly for y in a small disk around 0. The key is that each local factor $H_p(s, y)$ deviates from 1 by a quantity of order $p^{-2\,\mathrm{Re}(s)}$.

Lemma 3.3. Fix $\eta \in (0, 1/2)$ and $\delta \in (0, 1/2)$. Write $u := p^{-s}$ and denote the local factor by

$$H_p(s, y) := (1-u)^{2+2y} \left(1 + (1+y)\left((1-u)^{-2} - 1\right)\right).$$

Then for $|y| \le \eta$ and $\mathrm{Re}(s) \ge 1 - \delta$,

$$H_p(s, y) = 1 + O_{\eta,\delta}(u^2), \tag{3.5}$$

where the implied constant depends only on η and δ.

Proof. The condition $\mathrm{Re}(s) \ge 1 - \delta$ implies that, for every prime $p \ge 2$,

$$|u| = p^{-\mathrm{Re}(s)} \le p^{-(1-\delta)} \le 2^{-(1-\delta)} < 1,$$

since $\delta < 1/2$. In particular, $|u|$ is bounded away from 1, and the power series expansions

$$(1-u)^{-2} - 1 = 2u + 3u^2 + 4u^3 + \cdots = 2u + O_\delta(u^2),$$

$$(1-u)^{2+2y} = 1 - (2+2y)u + \binom{2+2y}{2} u^2 + \cdots = 1 - (2+2y)u + O_{\eta,\delta}(u^2),$$

converge absolutely, with remainders uniform for $|y| \le \eta$ and $\mathrm{Re}(s) \ge 1 - \delta$. Multiplying the expansions:

$$H_p(s, y) = \left(1 - (2+2y)u + O_{\eta,\delta}(u^2)\right)\left(1 + (1+y)(2u + O_\delta(u^2))\right).$$

The coefficient of u^1 on the right is $-(2+2y) + 2(1+y) = 0$. Hence $H_p(s, y) = 1 + O_{\eta,\delta}(u^2)$, proving (3.5). $\square$

Proposition 3.4. Fix $\eta \in (0, 1/2)$ and $\delta \in (0, 1/2)$, and let

$$\mathcal{R}_{\delta,\eta} := \{(s, y) \in \mathbb{C}^2 : \mathrm{Re}(s) \ge 1 - \delta,\ |y| \le \eta\}. \tag{3.6}$$

Then the partial products of the Euler product (3.3) converge absolutely and uniformly on $\mathcal{R}_{\delta,\eta}$ to a function $H(s, y)$. For every y with $|y| \le \eta$, the function $H(s, y)$ is holomorphic in s on the open strip $\mathrm{Re}(s) > 1 - \delta$. Moreover, $H(s, y)$ is jointly holomorphic in (s, y) on $\mathrm{int}(\mathcal{R}_{\delta,\eta})$, and H is uniformly bounded on $\mathcal{R}_{\delta,\eta}$.

Proof. By Lemma 3.3, for every $(s, y) \in \mathcal{R}_{\delta,\eta}$ and every prime p,

$$|H_p(s, y) - 1| \le C_{\eta,\delta}\, p^{-2(1-\delta)}, \tag{3.7}$$

where the constant $C_{\eta,\delta}$ depends only on η and δ. Since $\delta < 1/2$, we have $2(1 - \delta) > 1$, and the series $\sum_n n^{-2(1-\delta)}$ converges; in particular, $\sum_p p^{-2(1-\delta)}$ converges. By the Weierstrass M-test, the series

$$\sum_p |H_p(s, y) - 1|$$

converges uniformly on $\mathcal{R}_{\delta,\eta}$. Hence the infinite product criterion applies: if $\sum_n |u_n|$ converges uniformly, then the product $\prod_n (1 + u_n)$ converges absolutely and uniformly. Therefore the partial products $\prod_{p \le P} H_p(s, y)$ converge absolutely and uniformly on $\mathcal{R}_{\delta,\eta}$ to a limit, which we denote $H(s, y)$.

To deduce holomorphy in s, fix $y_0 \in \mathbb{C}$ with $|y_0| \le \eta$ and let $s_0 \in \mathbb{C}$ be any point with $\mathrm{Re}(s_0) > 1 - \delta$. Choose $r > 0$ small enough that $r < \mathrm{Re}(s_0) - (1 - \delta)$, and define

$$K := \overline{D}(s_0, r) = \{s \in \mathbb{C} : |s - s_0| \le r\}. \tag{3.8}$$

The set K is closed and bounded in $\mathbb{C}$, so by the Heine–Borel theorem it is compact. By the choice of r, every $s \in K$ satisfies

$$\mathrm{Re}(s) \ge \mathrm{Re}(s_0) - r > 1 - \delta,$$

and hence $K \times \{y_0\} \subset \mathcal{R}_{\delta,\eta}$. The uniform convergence of the partial products on $\mathcal{R}_{\delta,\eta}$ established above therefore restricts to uniform convergence of $\prod_{p \le P} H_p(s, y_0)$ on K as $P \to \infty$.

Each finite partial product $\prod_{p \le P} H_p(s, y_0)$ is holomorphic in s on the open disk $D(s_0, r) = \mathrm{int}(K)$, since each factor $H_p(\cdot, y_0)$ is. By the theorem on uniform limits of holomorphic functions (Weierstrass), the limit $H(\cdot, y_0)$ is holomorphic on $D(s_0, r)$, and in particular at s_0. Since s_0 with $\mathrm{Re}(s_0) > 1 - \delta$ was arbitrary, $H(\cdot, y_0)$ is holomorphic on the open strip $\mathrm{Re}(s) > 1 - \delta$. Since y_0 with $|y_0| \le \eta$ was also arbitrary, the holomorphy claim in s is proved.

For joint holomorphy, each finite partial product $\prod_{p \le P} H_p(s, y)$ is holomorphic in (s, y) on $\mathrm{int}(\mathcal{R}_{\delta,\eta})$, since each factor $H_p(s, y)$ is. The uniform convergence of the partial products on $\mathcal{R}_{\delta,\eta}$ established above restricts to uniform convergence on every compact subset of $\mathrm{int}(\mathcal{R}_{\delta,\eta})$, since $\mathrm{int}(\mathcal{R}_{\delta,\eta}) \subset \mathcal{R}_{\delta,\eta}$. By the theorem on uniform limits of holomorphic functions in two complex variables, the limit $H(s, y)$ is jointly holomorphic on $\mathrm{int}(\mathcal{R}_{\delta,\eta})$.

Uniform boundedness on $\mathcal{R}_{\delta,\eta}$ follows from (3.7):

$$|H(s, y)| \le \prod_p \bigl(1 + |H_p(s, y) - 1|\bigr) \le \exp\left(\sum_p |H_p(s, y) - 1|\right) \le \exp\left(C_{\eta,\delta} \sum_p p^{-2(1-\delta)}\right),$$

the right-hand side being finite and independent of $(s, y) \in \mathcal{R}_{\delta,\eta}$. □

Lemma 3.5. The function $H(s, y)$ satisfies

$$H(s, 0) \equiv 1 \quad \text{on } \mathrm{Re}(s) > 1/2. \tag{3.9}$$

In particular, $H(1, 0) = 1$.

Proof. On $\mathrm{Re}(s) > 1$, the factorization (3.4) specialized at $y = 0$ reads

$$F(s, 0) = \zeta(s)^2\, H(s, 0).$$

On the other hand, the Dirichlet series defining F gives

$$F(s,0) = \sum_{n\geq 1} \frac{d(n)}{n^s} = \zeta(s)^2 \quad \text{on } \operatorname{Re}(s) > 1.$$

Comparing the two expressions yields $\zeta(s)^2 H(s,0) = \zeta(s)^2$ on $\operatorname{Re}(s) > 1$. Since $\zeta(s) \neq 0$ on $\operatorname{Re}(s) > 1$ (which follows directly from the Euler product representation of ζ), we may divide both sides by $\zeta(s)^2$ to obtain

$$H(s,0) \equiv 1 \quad \text{on } \operatorname{Re}(s) > 1.$$

By Proposition 3.4, applied with any $\delta \in (0,1/2)$, the function $H(s,0)$ is holomorphic on $\operatorname{Re}(s) > 1-\delta$; letting $\delta \to 1/2$ shows that $H(s,0)$ is holomorphic on the connected open set $\operatorname{Re}(s) > 1/2$. The constant function 1 is also holomorphic there, and the two functions agree on the open subset $\operatorname{Re}(s) > 1$. By the identity theorem, they agree on all of $\operatorname{Re}(s) > 1/2$. In particular, $H(1,0) = 1$. □

4 Perron formula and contour deformation

To extract the asymptotic behavior of $T_j(x)$ from the Dirichlet series $F(s,y)$, we work with the generating partial sum

$$A(x,y) := \sum_{n\leq x} d(n)(1+y)^{\omega(n)}, \tag{4.1}$$

which satisfies the generating relation

$$A(x,y) = \sum_{j\geq 0} T_j(x) y^j. \tag{4.2}$$

Throughout this section, we initially fix one parameter:

- $\eta \in (0,1/2)$, controlling the size of the disk $|y| \leq \eta$.

After the contour depth δ' is fixed, we may shrink η if necessary for the contour estimates.

The fixed strip depth δ' and the truncation-dependent quantity λ_T will be chosen together in Section 4.3. This keeps the Euler factor estimates, the zero-free region, and the contour growth bound tied to the same strip parameter.

We also abbreviate $\alpha := 2+2y$. Thus the factorization (3.4) becomes $F(s,y) = \zeta(s)^\alpha H(s,y)$. For $|y| \leq \eta$ with $\eta < 1/2$, the exponent α ranges in a disk of radius $2\eta < 1$ around 2, so in particular α is bounded away from non-positive integers.

4.1 The truncated Perron formula

We quote the following standard form of the truncated Perron formula (see, e.g., []).

Theorem 4.1 (Truncated Perron formula)**.** Let $f(s) = \sum_{n\geq 1} a_n n^{-s}$ be a Dirichlet series with abscissa of absolute convergence σ_a. For real numbers $c > \sigma_a$, $x \geq 2$, and $T \geq 1$ with $x \notin \mathbb{Z}$,

$$\sum_{n\leq x} a_n = \frac{1}{2\pi i} \int_{c-iT}^{c+iT} f(s) \frac{x^s}{s}\, ds + R(x,T), \tag{4.3}$$

where the truncation error satisfies

$$R(x,T) \ll x^c \sum_{n\geq 1} \frac{|a_n|}{n^c} \frac{1}{1+T\left|\log \frac{x}{n}\right|}. \tag{4.4}$$

If $x = N$ is an integer, we do not apply the Perron formula directly at N. Instead, we apply it at

$$X := N + \frac{1}{2}.$$

Then $X \notin \mathbb{Z}$ and

$$A(N, y) = A(X, y),$$

because the integers satisfying $n \le N$ are exactly the integers satisfying $n \le N + 1/2$. Moreover,

$$X = N + O(1), \qquad \log X = \log N + O(1/N),$$

so every asymptotic estimate obtained for X transfers back to N. Thus it is enough in the Perron argument to work with non-integral x; the integer case follows from this replacement.

We apply Theorem 4.1 to the Dirichlet series $F(s, y) = \sum_{n \ge 1} d(n)(1+y)^{\omega(n)} n^{-s}$. For $|y| \le \eta$ with η sufficiently small, the series converges absolutely for $\mathrm{Re}(s) > 1$, so we may take any $c > 1$. Choose, for instance, $c = 1 + 1/\log x$, which is the standard choice to balance the main term size and the truncation error.

For $x \ge 2$ with $x \notin \mathbb{Z}$, $|y| \le \eta$, and $T \ge 1$, put $c = 1 + 1/\log x$. Theorem 4.1 gives

$$A(x, y) = \frac{1}{2\pi i} \int_{c-iT}^{c+iT} F(s, y) \frac{x^s}{s} \, ds + E_{\mathrm{Perron}}(x, T, y), \tag{4.5}$$

where

$$E_{\mathrm{Perron}}(x, T, y) \ll_\eta x^c \sum_{n \ge 1} \frac{d(n)(1+\eta)^{\omega(n)}}{n^c} \frac{1}{1 + T \left|\log \frac{x}{n}\right|}. \tag{4.6}$$

Indeed, applying (4.4) with $a_n = d(n)(1+y)^{\omega(n)}$ and using $|1+y| \le 1+\eta$ gives the displayed bound.

4.2 Analytic preliminaries

To control the contour integrals arising from the deformation in Section 4.3, we will need:

(a) a region in which $\zeta(s)$ does not vanish, so that the contour can avoid its zeros;

(b) a continuous branch of $\zeta(s)^{2+2y}$ on a neighborhood of the contour;

(c) a uniform bound on $|\zeta(s)|$ along vertical lines in the critical strip;

(d) a corresponding bound on $\arg \zeta(s)$ on the chosen branch.

We collect these analytic ingredients here, before constructing the contour itself in Section 4.3.

Zero-free region

For the contour deformation to be valid, we need a region in which $\zeta(s) \ne 0$. We use the following classical result of de la Vallée Poussin, quoted without proof; see, e.g., [].

Theorem 4.2 (de la Vallée Poussin zero-free region)**.** There exists an absolute constant $c_0 \in (0, 1/2)$ such that $\zeta(s) \ne 0$ on the open region

$$\mathcal{Z} := \left\{ s = \sigma + it \in \mathbb{C} : \sigma > 1 - \frac{c_0}{\log(|t| + 2)} \right\} \setminus \{1\}. \tag{4.7}$$

The boundary of $\mathcal{Z}$ is curved, and integrating along it would complicate the estimation of the resulting contour pieces. We will instead use a fixed vertical line lying strictly inside $\mathcal{Z}$ for all $|t| \le T$; the precise depth of this line is fixed in Section 4.3.

Continuous branch of $\zeta(s)^\alpha$

To make sense of $\zeta(s)^\alpha = \zeta(s)^{2+2y}$ on a neighborhood of the contour, we specify a branch on the region

$$\mathcal{D} := \mathcal{Z} \setminus \mathcal{S}, \tag{4.8}$$

where $\mathcal{S} := \{s \in \mathbb{C} : \Im(s) = 0,\ \mathrm{Re}(s) \le 1\}$ is the slit along the real axis to the left of $s = 1$. The slit $\mathcal{S}$ disconnects the segment $\mathcal{Z} \cap \mathcal{S}$ from the rest of $\mathcal{Z}$ from above and below, rendering $\mathcal{D}$ simply connected.[1]

On $\mathcal{D}$, both $\zeta(s)$ and $\zeta'(s)$ are holomorphic, and $\zeta(s) \neq 0$. Fix any real number $c_* > 1$ (for example $c_* = 2$); since $\zeta(c_*) > 0$, the principal logarithm $\log \zeta(c_*) \in \mathbb{R}$ is well-defined. Define

$$L(s) := \log \zeta(c_*) + \int_{c_*}^{s} \frac{\zeta'(\omega)}{\zeta(\omega)}\, d\omega \qquad (s \in \mathcal{D}), \tag{4.9}$$

where the integral is taken along any path in $\mathcal{D}$ from c_* to s. Since $\mathcal{D}$ is simply connected and ζ'/ζ is holomorphic on $\mathcal{D}$, the integral is path-independent, and L is a well-defined holomorphic function on $\mathcal{D}$.

We claim $e^{L(s)} = \zeta(s)$ on $\mathcal{D}$. Indeed, $L'(s) = \zeta'(s)/\zeta(s)$ on $\mathcal{D}$, so

$$\frac{d}{ds}\big(e^{-L(s)}\zeta(s)\big) = e^{-L(s)}\big(\zeta'(s) - L'(s)\zeta(s)\big) = 0.$$

Hence $e^{-L(s)}\zeta(s)$ is constant on $\mathcal{D}$, and evaluating at $s = c_*$ gives $e^{-L(c_*)}\zeta(c_*) = e^{-\log\zeta(c_*)}\zeta(c_*) = 1$. Therefore $e^{L(s)} = \zeta(s)$ throughout $\mathcal{D}$, so L serves as a continuous branch of $\log \zeta$ on $\mathcal{D}$.

We define

$$\zeta(s)^\alpha := e^{\alpha L(s)} \qquad (s \in \mathcal{D},\ \alpha \in \mathbb{C}). \tag{4.10}$$

This is single-valued and holomorphic in s on $\mathcal{D}$, and entire in α. At $\alpha = 2$, the definition reduces to $\zeta(s)^2$ in the usual sense.

Vertical growth of $\zeta(s)$

We use the following uniform bound on $|\zeta(s)|$, obtained by combining the convexity bound with the trivial estimate $\zeta(\sigma + it) \ll \log|t|$ for $\sigma \ge 1$ (see, e.g., [] for the convexity bound).

Theorem 4.3 (Uniform bound on $\zeta(s)$ in a vertical strip)**.** Fix $\delta \in (0, 1/2)$. For every $\varepsilon > 0$,

$$|\zeta(\sigma + it)| \ll_{\delta,\varepsilon} |t|^{\delta/2+\varepsilon} \quad \text{for } 1 - \delta \le \sigma \le 1 + \delta \text{ and } |t| \ge 2, \tag{4.11}$$

where the implied constant is independent of t.

Sketch. For $1 - \delta \le \sigma \le 1$, the convexity bound gives $|\zeta(\sigma + it)| \ll_\varepsilon |t|^{(1-\sigma)/2+\varepsilon}$. Taking the supremum of $(1-\sigma)/2$ over $\sigma \in [1-\delta, 1]$ yields $\delta/2$, so $|\zeta(\sigma + it)| \ll_\varepsilon |t|^{\delta/2+\varepsilon}$ on this subrange.

For $1 \le \sigma \le 1+\delta$, the standard estimate $|\zeta(\sigma+it)| \ll \log|t|$ holds, and $\log|t| \ll_\varepsilon |t|^\varepsilon \le |t|^{\delta/2+\varepsilon}$. Combining the two ranges gives (4.11). □

[1] Removing the slit from $s = 1$ to the boundary cuts the only possible loop around the puncture $s = 1$, just as $\mathbb{C} \setminus (-\infty, 0]$ is the standard simply connected branch domain for the logarithm. This is the topological input needed for path independence in the definition of $L(s)$.

Bound on $\arg \zeta$

We also use the following standard estimate on the branch determined by (4.9); see, e.g., [].

Theorem 4.4 (Bound on $\arg \zeta$)**.** There exists an absolute constant $C_{\arg} > 0$ such that, for every $s = \sigma + it \in \mathcal{D}$ with $|t| \geq 2$,

$$|\arg \zeta(s)| \leq C_{\arg} \log |t|, \tag{4.12}$$

where the branch of $\arg \zeta$ is the one determined by the logarithm L on $\mathcal{D}$.

4.3 Contour decomposition

We now introduce the parameters governing the contour deformation and construct the deformed contour.

Depth of the deformation

Fix once and for all an absolute constant

$$\delta' \in \left(\frac{c_0}{2 \log 3}, \frac{c_0}{2} \right), \tag{4.13}$$

where c_0 is the constant of Theorem 4.2; the interval is non-empty since $\log 3 > 1$. This range of δ' is chosen so that the deformed contour remains in the fixed strip used in Section 4.4.

For each truncation height $T \geq 1$, define

$$\lambda_T := \frac{c_0}{2 \log(T+2)}. \tag{4.14}$$

By construction, $\lambda_T \to 0$ as $T \to \infty$, and for every $T \geq 1$,

$$\lambda_T \leq \frac{c_0}{2 \log 3} < \delta', \tag{4.15}$$

so the depth $\sigma_0(T) := 1 - \lambda_T$ of the deformed contour satisfies $\sigma_0(T) > 1 - \delta'$. Consequently, the contour $\mathcal{C}(T, \rho)$ defined below lies in $\mathcal{R}_{\delta',\eta}$ for every $T \geq 1$ and every $|y| \leq \eta$. In particular, Proposition 3.4 applies on the contour with $\delta = \delta'$.

Finally, the choice $c = 1 + 1/\log x$ in the Perron representation (4.5) satisfies $c \leq 1 + \delta'$ for all x sufficiently large, ensuring that the abscissa c lies inside the strip $1 - \delta' \leq \sigma \leq 1 + \delta'$ used in the contour estimates. We assume this hereafter.

The deformed contour

Fix a truncation height $T \geq 1$, and set

$$\sigma_0 := \sigma_0(T) := 1 - \lambda_T. \tag{4.16}$$

Choose a parameter $\rho > 0$ with $\rho < \lambda_T$; its precise choice as a function of x will be made in Section 5. The notation $i0^{\pm}$ below denotes boundary values: the two Hankel arms are first taken at heights $\pm\varepsilon$ with $\varepsilon > 0$, and the limit $\varepsilon \to 0^+$ is taken after the corresponding integrals are formed. Thus the approximating contours lie in the slit domain $\mathcal{D}$ before the boundary value limit is taken.

We deform the original Perron segment $[c - iT, c + iT]$ leftward, around the singularity at $s = 1$, into a contour $\mathcal{C} = \mathcal{C}(T, \rho)$ traversed from $c - iT$ to $c + iT$. It consists of the following five pieces:

(1) $\mathcal{C}_{\mathrm{bot}}$: the horizontal segment from $c - iT$ to $\sigma_0 - iT$.

(2) $\mathcal{C}^-_{\mathrm{left}}$: the vertical segment from $\sigma_0 - iT$ to $\sigma_0 + i0^-$.

(3) $\mathcal{H}_{\mathrm{loc}}$: a local Hankel contour around $s = 1$, consisting of

(3a) the lower arm from $\sigma_0 + i0^-$ to $1 - \rho + i0^-$;

(3b) the circle $|s - 1| = \rho$, traversed from $1 - \rho + i0^-$ to $1 - \rho + i0^+$ counterclockwise through the right side of $s = 1$;

(3c) the upper arm from $1 - \rho + i0^+$ to $\sigma_0 + i0^+$.

(4) $\mathcal{C}^+_{\mathrm{left}}$: the vertical segment from $\sigma_0 + i0^+$ to $\sigma_0 + iT$.

(5) $\mathcal{C}_{\mathrm{top}}$: the horizontal segment from $\sigma_0 + iT$ to $c + iT$.

The five pieces are joined consecutively, so $\mathcal{C}$ has the same initial and terminal points as the original Perron contour. The Hankel arms are understood as boundary values along the slit $\mathcal{S}$; in the local representation near $s = 1$, their discontinuity is carried by the factor $(s-1)^{-2-2y}$ and produces the local contribution I_H.

Verification that the contour avoids ζ-zeros

We claim that $\mathcal{C} \setminus \{1\} \subset \mathcal{Z}$, where $\mathcal{Z}$ is the zero-free region of Theorem 4.2. For the horizontal pieces $\mathcal{C}_{\mathrm{bot}}$ and $\mathcal{C}_{\mathrm{top}}$,

$$\mathrm{Re}(s) \geq \sigma_0(T) = 1 - \frac{c_0}{2\log(T+2)} > 1 - \frac{c_0}{\log(|\Im s| + 2)}. \tag{4.17}$$

For the vertical pieces $\mathcal{C}^{\pm}_{\mathrm{left}}$,

$$\mathrm{Re}(s) = \sigma_0(T) = 1 - \frac{c_0}{2\log(T+2)} > 1 - \frac{c_0}{\log(|\Im s| + 2)}. \tag{4.18}$$

For the local Hankel contour $\mathcal{H}_{\mathrm{loc}}$, the two horizontal arms (3a) and (3c) satisfy $\mathrm{Re}(s) \geq \sigma_0(T)$ and $|\Im s| \leq \rho$, hence are covered by the same estimate as the vertical pieces. For the small circle (3b),

$$\mathrm{Re}(s) \geq 1 - \rho > 1 - \lambda_T = 1 - \frac{c_0}{2\log(T+2)} > 1 - \frac{c_0}{\log(|\Im s| + 2)}. \tag{4.19}$$

Therefore $\mathcal{C} \setminus \{1\} \subset \mathcal{Z}$. Combined with the boundary-value convention for the Hankel arms, the approximating contours with heights $\pm\varepsilon$ are contained in $\mathcal{D}$ away from $s = 1$. The integrand $F(s, y)x^s/s$ is therefore well-defined and holomorphic on a neighborhood of each approximating contour, and the contour $\mathcal{C}$ is understood as the corresponding boundary-value limit.

Cauchy's theorem

For $\varepsilon > 0$, replace the two Hankel arms along the slit by parallel paths at heights $-\varepsilon$ and $+\varepsilon$, and denote the resulting deformed contour by $\mathcal{C}_\varepsilon$. The Perron contour and $\mathcal{C}_\varepsilon$ share the same endpoints $c \pm iT$. Together, the Perron contour traversed from $c - iT$ to $c + iT$ and $\mathcal{C}_\varepsilon$ traversed in the reverse direction form a closed loop Γ_ε, oriented counterclockwise. The keyhole formed by the approximating Hankel contour and the slit $\mathcal{S}$ excludes the point $s = 1$ from the interior of Γ_ε, so the interior of Γ_ε is contained in the rectangle

$$\mathcal{R}_\Gamma := \{s = \sigma + it : \sigma_0(T) \leq \sigma \leq c,\ |t| \leq T\} \setminus \{1\}, \tag{4.20}$$

with the slit further excluded. For every $s \in \mathcal{R}_\Gamma$,

$$\operatorname{Re}(s) \geq \sigma_0(T) = 1 - \frac{c_0}{2\log(T+2)} > 1 - \frac{c_0}{\log(|\Im s|+2)}, \tag{4.21}$$

so $\mathcal{R}_\Gamma \subset \mathcal{Z}$. Combined with the on-contour verifications (4.17)–(4.19), this shows that Γ_ε together with its interior, with the slit removed, is contained in the slit domain $\mathcal{D}$. Hence $F(s,y)x^s/s$ is holomorphic on an open neighborhood of Γ_ε and its interior. By Cauchy's theorem,

$$\oint_{\Gamma_\varepsilon} F(s,y)\frac{x^s}{s}\,ds = 0, \tag{4.22}$$

which yields

$$\int_{c-iT}^{c+iT} F(s,y)\frac{x^s}{s}\,ds = \int_{\mathcal{C}_\varepsilon} F(s,y)\frac{x^s}{s}\,ds.$$

Letting $\varepsilon \to 0^+$ gives the boundary-value contour $\mathcal{C}$ and hence

$$\int_{c-iT}^{c+iT} F(s,y)\frac{x^s}{s}\,ds = \int_{\mathcal{C}} F(s,y)\frac{x^s}{s}\,ds. \tag{4.23}$$

Decomposition of $A(x,y)$

For non-integral x, combining (4.23) with the Perron representation (4.5), we obtain

$$A(x,y) = I_H + I_{\text{top}} + I_{\text{bot}} + I_{\text{left}} + E_{\text{Perron}}(x,T,y), \tag{4.24}$$

where, for $* \in \{\text{top, bot, left}\}$,

$$I_* := \frac{1}{2\pi i}\int_{\mathcal{C}_*} F(s,y)\frac{x^s}{s}\,ds, \tag{4.25}$$

with $\mathcal{C}_{\text{left}} := \mathcal{C}_{\text{left}}^- \cup \mathcal{C}_{\text{left}}^+$, and

$$I_H := \frac{1}{2\pi i}\int_{\mathcal{H}_{\text{loc}}} F(s,y)\frac{x^s}{s}\,ds. \tag{4.26}$$

Writing $\mathcal{H}_{\text{loc}} = \mathcal{H}_{\text{loc}}^- \cup \mathcal{H}_{\text{loc}}^\circ \cup \mathcal{H}_{\text{loc}}^+$ for the lower arm, the small circle, and the upper arm, respectively, fixes the geometric decomposition of the local contour. The lower arm is oriented from $\sigma_0 + i0^-$ to $1-\rho+i0^-$, the small circle is oriented counterclockwise from $1-\rho+i0^-$ to $1-\rho+i0^+$ through the right side of $s=1$, and the upper arm is oriented from $1-\rho+i0^+$ to σ_0+i0^+.

The local Hankel contribution I_H contains the main asymptotic behavior and will be evaluated in Section 5. The remaining contributions $I_{\text{top}}, I_{\text{bot}}, I_{\text{left}}$ are estimated in Section 4.4 and shown to be of lower order than the main term, after a suitable choice of $T = T(x)$.

4.4 Estimates of the contour pieces

In this section, we estimate the three nonlocal contour contributions $I_{\text{top}}, I_{\text{bot}}, I_{\text{left}}$ appearing in the decomposition (4.24), and combine them with the Perron truncation error E_{Perron} to obtain a single error bound. The local Hankel contribution I_H is deferred to Section 5.

We first collect the two bounds for $F(s,y)$ used below.

Lemma 4.5 (Bounds for $F(s,y)$ on the deformed contour)**.** Let $0 < \eta < 1/2$, and let δ' be fixed as in (4.13). Let $\varepsilon > 0$ and put $\alpha = 2+2y$. Define

$$B := B(\eta,\delta',\varepsilon) := (2+2\eta)\left(\tfrac{\delta'}{2}+\varepsilon\right) + 2\eta C_{\text{arg}}. \tag{4.27}$$

Then the following estimates hold uniformly for $|y| \leq \eta$.

1. If $s = \sigma + it \in \mathcal{D}$, $|t| \geq 2$, and $1 - \delta' \leq \sigma \leq 1 + \delta'$, then

$$|F(s, y)| \ll_{\eta,\delta',\varepsilon} |t|^B. \tag{4.28}$$

2. If $T \geq 1$, $s = 1 - \lambda_T + it$, and $|t| \leq 2$, then

$$|F(s, y)| \ll_{\eta,\delta'} \lambda_T^{-2-2\eta} \ll_{\eta,\delta'} (\log(T + 2))^{2+2\eta}. \tag{4.29}$$

Proof. For the first estimate, let $s = \sigma + it \in \mathcal{D}$ with $|t| \geq 2$ and $1 - \delta' \leq \sigma \leq 1 + \delta'$. On $\mathcal{D}$, the chosen branch gives

$$F(s, y) = \zeta(s)^\alpha H(s, y), \qquad \zeta(s)^\alpha = \exp(\alpha L(s)).$$

Writing $L(s) = \log |\zeta(s)| + i \arg \zeta(s)$ on this branch,

$$|\zeta(s)^\alpha| = |\zeta(s)|^{\mathrm{Re}(\alpha)} \exp\bigl(-\Im(\alpha) \arg \zeta(s)\bigr).$$

Since $|y| \leq \eta$, we have $\mathrm{Re}(\alpha) \leq 2 + 2\eta$ and $|\Im(\alpha)| \leq 2\eta$. Hence

$$|F(s, y)| \leq |H(s, y)|(1 + |\zeta(s)|)^{2+2\eta} \exp\bigl(2\eta |\arg \zeta(s)|\bigr).$$

Here the factor $1 + |\zeta(s)|$ avoids any lower-bound assumption on $|\zeta(s)|$. Proposition 3.4, applied with $\delta = \delta'$, gives $|H(s, y)| \ll_{\eta,\delta'} 1$. Moreover, Theorem 4.3 gives

$$|\zeta(s)| \ll_{\delta',\varepsilon} |t|^{\delta'/2+\varepsilon},$$

and Theorem 4.4 gives

$$|\arg \zeta(s)| \leq C_{\arg} \log |t|.$$

Since $|t| \geq 2$, these estimates imply

$$|F(s, y)| \ll_{\eta,\delta',\varepsilon} |t|^{(2+2\eta)(\delta'/2+\varepsilon)+2\eta C_{\arg}} = |t|^B,$$

which proves (4.28).

We now prove the low-height estimate. Let

$$q(s) := (s - 1)\zeta(s), \qquad q(1) := 1.$$

Then q is holomorphic at $s = 1$. Put

$$\lambda_0 := \frac{c_0}{2 \log 3}, \qquad K := \{s = \sigma + it : 1 - \lambda_0 \leq \sigma \leq 1,\ |t| \leq 2\}.$$

For $|t| \leq 2$ we have $\log(|t| + 2) \leq \log 4 < 2 \log 3$, and hence

$$\lambda_0 = \frac{c_0}{2 \log 3} < \frac{c_0}{\log(|t| + 2)}.$$

Thus $K \setminus \{1\}$ lies in the zero-free region of Theorem 4.2. After setting $q(1) = 1$, the function q is holomorphic and non-vanishing on K. Since K is compact, there is an open neighborhood U of K on which q is non-vanishing. Choose a holomorphic branch

$$M(s) = \log q(s)$$

on U, compatible with $M(s) = L(s) + \mathrm{Log}(s-1)$ on $U \cap \mathcal{D}$. Let

$$M_K := \sup_{s\in K} |M(s)| < \infty.$$

For $|y| \le \eta$,

$$|q(s)^\alpha| = |\exp(\alpha M(s))| \le \exp(|\alpha|\,|M(s)|) \le \exp((2+2\eta)M_K) \qquad (s \in K).$$

Define

$$G(s,y) := q(s)^\alpha H(s,y). \tag{4.30}$$

Then, on the chosen branch near the slit,

$$F(s,y) = G(s,y)(s-1)^{-\alpha}. \tag{4.31}$$

Since $K \subset \{s : \mathrm{Re}(s) \ge 1-\delta'\}$, combining the preceding bound for $q(s)^\alpha$ with Proposition 3.4, again with $\delta = \delta'$, gives

$$|G(s,y)| = |q(s)^\alpha H(s,y)| \ll_{\eta,\delta'} 1 \qquad (s \in K,\ |y| \le \eta).$$

On the left vertical line $s = 1 - \lambda_T + it$ with $|t| \le 2$, we use (4.31). For either boundary value of the logarithm along the slit, $|\mathrm{Arg}(s-1)| \le \pi$, so

$$|(s-1)^{-\alpha}| = |s-1|^{-\mathrm{Re}(\alpha)} \exp\big(\Im(\alpha)\,\mathrm{Arg}(s-1)\big) \ll_\eta |s-1|^{-\mathrm{Re}(\alpha)}.$$

Since $|s-1| \ge \lambda_T$, if $|s-1| \le 1$ then

$$|s-1|^{-\mathrm{Re}(\alpha)} \le |s-1|^{-2-2\eta} \le \lambda_T^{-2-2\eta}.$$

If $|s-1| \ge 1$, then $\mathrm{Re}(\alpha) \ge 2-2\eta > 0$, so $|s-1|^{-\mathrm{Re}(\alpha)} \le 1 \le \lambda_T^{-2-2\eta}$. Hence

$$|F(1-\lambda_T+it, y)| \ll_{\eta,\delta'} \lambda_T^{-2-2\eta}.$$

The definition (4.14) gives

$$\lambda_T^{-2-2\eta} = \left(\frac{2\log(T+2)}{c_0}\right)^{2+2\eta} \ll_{\eta,\delta'} (\log(T+2))^{2+2\eta},$$

which proves (4.29). □

Since

$$B(\eta,\delta',\varepsilon) = (2+2\eta)\left(\tfrac{\delta'}{2}+\varepsilon\right) + 2\eta C_{\mathrm{arg}} \longrightarrow \delta' \qquad (\eta,\varepsilon \to 0^+),$$

and $\delta' < c_0/2$, we may choose $\eta > 0$ and $\varepsilon > 0$ sufficiently small so that

$$B < c_0/2. \tag{4.32}$$

This choice is fixed throughout the contour estimates.

Nonlocal contour pieces

Lemma 4.6 (Bound for the horizontal contour pieces)**.** For all sufficiently large x and every $T \ge 2$,

$$|I_{\mathrm{top}}| + |I_{\mathrm{bot}}| \ll_{\eta,\varepsilon} x^c \, T^{B-1}, \tag{4.33}$$

uniformly in $|y| \le \eta$.

Proof. On $\mathcal{C}_{\mathrm{top}}$, parameterize $s = \sigma + iT$ with $\sigma_0(T) \le \sigma \le c$. Then

$$\begin{aligned}
|I_{\mathrm{top}}| &= \left| \frac{1}{2\pi i} \int_{\mathcal{C}_{\mathrm{top}}} F(s,y) \frac{x^s}{s} \, ds \right| \\
&\le \frac{1}{2\pi} \int_{\sigma_0(T)}^{c} \left| F(\sigma + iT, y) \frac{x^{\sigma+iT}}{\sigma + iT} \right| d\sigma \\
&\le \frac{c - \sigma_0(T)}{2\pi} \sup_{\sigma_0(T) \le \sigma \le c} \left| F(\sigma + iT, y) \frac{x^{\sigma+iT}}{\sigma + iT} \right|.
\end{aligned}$$

On this segment, $|x^{\sigma+iT}| = x^\sigma \le x^c$ and $|\sigma + iT| \ge T$. Also $\sigma_0(T) \ge 1 - \delta'$ by (4.15), while $c \le 1 + \delta'$ for all sufficiently large x. Hence Lemma 4.5 gives

$$\sup_{\sigma_0(T) \le \sigma \le c} |F(\sigma + iT, y)| \ll_{\eta,\varepsilon} T^B.$$

Since

$$c - \sigma_0(T) = \frac{1}{\log x} + \lambda_T = \frac{1}{\log x} + \frac{c_0}{2\log(T+2)} \ll 1,$$

we obtain

$$\begin{aligned}
|I_{\mathrm{top}}| &\ll_{\eta,\varepsilon} \frac{x^c}{T} \sup_{\sigma_0(T) \le \sigma \le c} |F(\sigma + iT, y)| \\
&\ll_{\eta,\varepsilon} x^c T^{B-1}.
\end{aligned}$$

The same computation on $\mathcal{C}_{\mathrm{bot}}$, with $s = \sigma - iT$, gives

$$|I_{\mathrm{bot}}| \ll_{\eta,\varepsilon} x^c T^{B-1}.$$

This proves (4.33). $\square$

Lemma 4.7 (Bound for the left vertical contour piece)**.** For every $T \ge 2$,

$$|I_{\mathrm{left}}| \ll_{\eta,\varepsilon} x^{1-\lambda_T} \, T^B, \tag{4.34}$$

uniformly in $|y| \le \eta$.

Proof. The contour $\mathcal{C}_{\mathrm{left}} = \mathcal{C}_{\mathrm{left}}^- \cup \mathcal{C}_{\mathrm{left}}^+$ is estimated by parameterizing the two boundary-value pieces as $s = \sigma_0(T) + it$, with $-T \le t < 0$ on $\mathcal{C}_{\mathrm{left}}^-$ and $0 < t \le T$ on $\mathcal{C}_{\mathrm{left}}^+$. Using $\sigma_0(T) = 1 - \lambda_T$, we first write

$$\begin{aligned}
|I_{\mathrm{left}}| &= \left| \frac{1}{2\pi i} \int_{\mathcal{C}_{\mathrm{left}}} F(s,y) \frac{x^s}{s} \, ds \right| \\
&\le \frac{1}{2\pi} \left(\int_{-T}^{0^-} \left| F(\sigma_0(T) + it, y) \frac{x^{\sigma_0(T)+it}}{\sigma_0(T) + it} \right| dt + \int_{0^+}^{T} \left| F(\sigma_0(T) + it, y) \frac{x^{\sigma_0(T)+it}}{\sigma_0(T) + it} \right| dt \right) \\
&= \frac{x^{1-\lambda_T}}{2\pi} \left(\int_{-T}^{0^-} \left| \frac{F(\sigma_0(T) + it, y)}{\sigma_0(T) + it} \right| dt + \int_{0^+}^{T} \left| \frac{F(\sigma_0(T) + it, y)}{\sigma_0(T) + it} \right| dt \right).
\end{aligned}$$

Since $\sigma_0(T) > 1-\delta' > 1/2$, we have $|\sigma_0(T)+it| \gg 1$ for $|t| \le 2$ and $|\sigma_0(T)+it| \gg |t|$ for $2 \le |t| \le T$. Hence

$$|I_{\text{left}}| \ll x^{1-\lambda_T} \left(\int_{|t|\le 2} |F(\sigma_0(T)+it,y)|\,dt + \int_{2\le|t|\le T} \frac{|F(\sigma_0(T)+it,y)|}{|t|}\,dt \right).$$

On $|t| \le 2$, Lemma 4.5 gives

$$|F(\sigma_0(T)+it,y)| \ll_{\eta,\delta'} (\log(T+2))^{2+2\eta}.$$

On $2 \le |t| \le T$, the point $\sigma_0(T)+it$ lies in $\mathcal{D}$ as a boundary-value limit and satisfies $1-\delta' \le \sigma_0(T) \le 1+\delta'$, so Lemma 4.5 gives

$$|F(\sigma_0(T)+it,y)| \ll_{\eta,\varepsilon} |t|^B.$$

Therefore

$$\begin{aligned} |I_{\text{left}}| &\ll_{\eta,\delta',\varepsilon} x^{1-\lambda_T} \left((\log(T+2))^{2+2\eta} + \int_{2\le|t|\le T} |t|^{B-1}\,dt \right) \\ &\ll_{\eta,\delta',\varepsilon} x^{1-\lambda_T} \left((\log(T+2))^{2+2\eta} + 2\int_2^T t^{B-1}\,dt \right). \end{aligned}$$

Since $B > 0$,

$$2\int_2^T t^{B-1}\,dt = \frac{2}{B}\big(T^B - 2^B\big) \ll_B T^B.$$

Also $(\log(T+2))^{2+2\eta} \ll_{\eta,B} T^B$ for $T \ge 2$. Combining these estimates gives

$$|I_{\text{left}}| \ll_{\eta,\delta',\varepsilon} x^{1-\lambda_T} T^B.$$

This proves (4.34). □

Choice of T and the combined error

We choose

$$T = T(x) := \exp\big(\sqrt{\log x}\big), \tag{4.35}$$

the standard Selberg–Delange truncation height adapted to the de la Vallée Poussin zero-free region. With this choice, $\log T = \sqrt{\log x}$, and $\lambda_T = c_0/(2\sqrt{\log x} + O(1))$, so $\lambda_T < \delta'$ holds automatically. Moreover, $c = 1 + 1/\log x \le 1+\delta'$ for all x sufficiently large, so the abscissa c lies inside the strip used in the growth estimates. We have

$$x^{-\lambda_T} = \exp(-\lambda_T \log x) = \exp\left(-\tfrac{c_0}{2}\sqrt{\log x}\cdot(1+o(1))\right). \tag{4.36}$$

Lemma 4.8 (Short interval bound for d_K)**.** For each fixed integer $K \ge 1$, there exists a constant $C_K > 0$ such that, for all sufficiently large X, if

$$C_K X^{1-1/(2K)} \le H \le X,$$

then

$$\sum_{X<n\le X+H} d_K(n) \ll_K H(\log X)^{K-1}.$$

Here $d_K(n)$ is defined by

$$\zeta(s)^K = \sum_{n\ge 1} d_K(n) n^{-s} \qquad (\mathrm{Re}(s) > 1),$$

or equivalently as the number of ordered K-tuples of positive integers whose product is n.

Proof. We prove the assertion by induction on K. For $K = 1$, we have $d_1(n) = 1$, so the estimate is immediate.

Assume $K \geq 2$ and that the result is known for $K - 1$. Put

$$D_K(X,H) := \sum_{X<n\leq X+H} d_K(n).$$

This counts ordered K-tuples $(m_1, \ldots, m_K)$ satisfying

$$X < m_1 \cdots m_K \leq X + H.$$

Since $H \leq X$, the product is at most $2X$, so at least one factor is at most $(2X)^{1/K}$. By the union bound over the K possible positions of such a factor,

$$D_K(X,H) \leq K \sum_{m\leq(2X)^{1/K}} \sum_{X/m<r\leq(X+H)/m} d_{K-1}(r).$$

Let $Y = X/m$ and $L = H/m$. Choose C_K large enough, in terms of K and C_{K-1}, so that whenever $H \geq C_K X^{1-1/(2K)}$ and $m \leq (2X)^{1/K}$, we have

$$L \geq C_{K-1} Y^{1-1/(2(K-1))}.$$

Also $L \leq Y$, since $H \leq X$. Hence the induction hypothesis applies to the inner sum and gives

$$\sum_{X/m<r\leq(X+H)/m} d_{K-1}(r) \ll_K \frac{H}{m}(\log X)^{K-2}.$$

Therefore

$$D_K(X,H) \ll_K H(\log X)^{K-2} \sum_{m\leq(2X)^{1/K}} \frac{1}{m} \ll_K H(\log X)^{K-1}.$$

$\square$

Lemma 4.9 (Perron truncation error)**.** Let $T = T(x) = \exp(\sqrt{\log x})$ and $c = 1 + 1/\log x$. For $x \geq 2$ with $x \notin \mathbb{Z}$ and $|y| \leq \eta$, there exists a fixed integer $K = K(\eta)$ such that

$$E_{\text{Perron}}(x,T,y) \ll_\eta \frac{x}{T}(\log x)^K. \tag{4.37}$$

Proof. Set $M = 1 + \eta$ and choose an integer $K = K(\eta)$ with $K \geq 2M$ and $K \geq 2$. We first show that

$$d(n)M^{\omega(n)} \leq d_K(n) \qquad (n \geq 1).$$

For a prime power p^a with $a \geq 1$,

$$d(p^a)M^{\omega(p^a)} = (a+1)M, \qquad d_K(p^a) = \binom{a+K-1}{K-1}.$$

The quantity

$$\frac{1}{a+1}\binom{a+K-1}{K-1}$$

is increasing in a for $K \geq 2$, and at $a = 1$ it equals $K/2$. Thus $(a+1)M \leq d_K(p^a)$ for every $a \geq 1$. Extending multiplicatively gives the claimed coefficient domination.

Using (4.6), we obtain

$$E_{\mathrm{Perron}}(x,T,y) \ll_\eta x^c \sum_{n\ge1} \frac{d_K(n)}{n^c} \frac{1}{1+T\left|\log\frac{x}{n}\right|}.$$

We split the sum into the ranges $n \le x/2$, $x/2 < n < 2x$, and $n \ge 2x$.

If $n \le x/2$ or $n \ge 2x$, then $|\log(x/n)| \ge \log 2$, and hence the weight is $\ll 1/T$. The contribution of these two far ranges is therefore

$$\ll \frac{x^c}{T} \sum_{n\ge1} \frac{d_K(n)}{n^c} = \frac{x^c}{T}\zeta(c)^K.$$

Since $c = 1 + 1/\log x$, we have $x^c \asymp x$ and $\zeta(c) \ll \log x$, so the far contribution is

$$\ll_K \frac{x}{T}(\log x)^K.$$

It remains to estimate the central range $x/2 < n < 2x$. There $x^c n^{-c} = (x/n)^c \ll 1$, so it is enough to bound

$$\sum_{x/2<n<2x} d_K(n) \frac{1}{1+T\left|\log\frac{x}{n}\right|}.$$

Put $H = x/T$. Cover the central range by the right intervals

$$I_r^+ = (x + rH,\ x + (r+1)H]$$

and the left intervals

$$I_r^- = [x - (r+1)H,\ x - rH),$$

with $r = 0, 1, 2, \ldots$ and only $O(T)$ intervals needed. For $n \in I_r^\pm$,

$$\frac{1}{1+T\left|\log\frac{x}{n}\right|} \ll \frac{1}{1+r}.$$

For $r = 0$ this is trivial. For $r \ge 1$, since $x/2 < n < 2x$, we have

$$\left|\log\frac{x}{n}\right| \asymp \frac{|n-x|}{x} \asymp \frac{rH}{x} = \frac{r}{T}.$$

Because $T = \exp(\sqrt{\log x})$, the length $H = x\exp(-\sqrt{\log x})$ is larger than $C_K x^{1-1/(2K)}$ for all sufficiently large x. Each interval lies in $x/2 < n < 2x$, so Lemma 4.8 applies with a starting point comparable to x, and gives

$$\sum_{n\in I_r^\pm} d_K(n) \ll_K H(\log x)^{K-1}.$$

Therefore the central contribution is

$$\ll_K H(\log x)^{K-1} \sum_{r\ll T} \frac{1}{1+r} \ll_K \frac{x}{T}(\log x)^{K-1}\log T.$$

Since $\log T = \sqrt{\log x} \le \log x$ for sufficiently large x, this is

$$\ll_K \frac{x}{T}(\log x)^K.$$

Combining the far and central estimates proves (4.37). □

Proposition 4.10. With $T = T(x) = \exp(\sqrt{\log x})$, there exists a constant $c_1 > 0$ (depending on η, ε but not on x) such that

$$E_{\mathrm{Perron}}(x, T, y) + |I_{\mathrm{top}}| + |I_{\mathrm{bot}}| + |I_{\mathrm{left}}| \ll_{\eta,\varepsilon} x \exp\bigl(-c_1 \sqrt{\log x}\bigr), \tag{4.38}$$

uniformly in $|y| \le \eta$, for all sufficiently large $x \notin \mathbb{Z}$.

Proof. We bound each term separately and take the worst.

Perron error. By Lemma 4.9 with $T = \exp(\sqrt{\log x})$,

$$E_{\mathrm{Perron}} \ll_\eta x \exp(-\sqrt{\log x})(\log x)^K.$$

Since $(\log x)^K = \exp(K \log\log x)$ and, for sufficiently large x,

$$K \log\log x \le \frac{1}{2}\sqrt{\log x},$$

we have

$$E_{\mathrm{Perron}} \ll_\eta x \exp\left(-\frac{1}{2}\sqrt{\log x}\right).$$

Horizontal pieces. With $c = 1 + 1/\log x$ so that $x^c = e \cdot x$, by Lemma 4.6,

$$|I_{\mathrm{top}}| + |I_{\mathrm{bot}}| \ll_{\eta,\varepsilon} x \cdot T^{B-1} = x \cdot \exp\bigl((B-1)\sqrt{\log x}\bigr).$$

Since $B < 1$, this is $\ll x \exp(-(1-B)\sqrt{\log x})$.

Vertical piece. By Lemma 4.7 and (4.36),

$$|I_{\mathrm{left}}| \ll_{\eta,\varepsilon} x \cdot x^{-\lambda_T} \cdot T^B = x \cdot \exp\bigl(-\tfrac{c_0}{2}\sqrt{\log x} + B\sqrt{\log x}\bigr)(1 + o(1)).$$

By the choice (4.13) of δ' together with (4.32), we have $B < c_0/2$. Hence $|I_{\mathrm{left}}| \ll x \exp(-(c_0/2 - B)\sqrt{\log x})$.

Set

$$c_1 := \frac{1}{2} \min\left(\frac{1}{2},\ 1 - B,\ \frac{c_0}{2} - B\right) > 0.$$

For all sufficiently large x, the preceding three estimates are bounded by the right-hand side of (4.38). ∎

The right-hand side of (4.38) is exponentially smaller than the local contribution on the $A(x, y)$ level. The local Hankel contribution I_H, analyzed in Section 5, contains the main term from which the asymptotic formula for each fixed $T_j(x)$ will later be extracted.

5 Local Hankel analysis

The purpose of this section is to analyze the local Hankel contribution I_H around the singular point $s = 1$. Recall that

$$F(s, y) = \zeta(s)^{2+2y} H(s, y).$$

The singular part is carried by the power of $\zeta(s)$; the remaining factor should be made regular at $s = 1$ before the local Hankel integral is evaluated.

5.1 The local regular factor

We keep the abbreviation

$$\alpha = 2 + 2y.$$

Define

$$q(s) := (s-1)\zeta(s), \qquad q(1) := 1. \tag{5.1}$$

The pole of $\zeta(s)$ at $s = 1$ is simple with residue 1, so q is holomorphic near $s = 1$ after filling in the removable singularity. After shrinking the neighborhood of $s = 1$ if necessary, we may assume that $q(s) \neq 0$ there. We then choose a single branch of $\log q(s)$ in this neighborhood, normalized by

$$\log q(1) = 0.$$

For s in this neighborhood and y near 0, define

$$G(s, y) := q(s)^{2+2y} H(s, y), \qquad q(s)^{2+2y} := \exp\big((2 + 2y) \log q(s)\big). \tag{5.2}$$

On the corresponding punctured slit neighborhood of $s = 1$, the branch choices from Section 4 give

$$F(s, y) = G(s, y)(s-1)^{-2-2y}. \tag{5.3}$$

Here the branch of $(s-1)^{-2-2y}$ is the local Hankel branch induced by the contour and branch conventions already fixed in Section 4.3.

Lemma 5.1 (Analyticity and boundedness of the local factor)**.** Fix $0 < \eta < 1/2$ and $0 < \delta < 1/2$. There exist $r \in (0, \delta)$ and $C_{\eta,\delta} > 0$ such that $G(s, y)$ is holomorphic in (s, y) for $|s-1| < r$ and $|y| < \eta$, and

$$|G(s, y)| \leq C_{\eta,\delta}$$

uniformly for $|s-1| \leq r$ and $|y| \leq \eta$. Moreover, on the punctured slit neighborhood

$$0 < |s-1| < r, \qquad s \notin \mathcal{S},$$

we have the local factorization

$$F(s, y) = G(s, y)(s-1)^{-2-2y}.$$

Proof. Apply Proposition 3.4 with this fixed δ. It gives that the function $H(s, y)$ is holomorphic in (s, y) on the interior of $\mathcal{R}_{\delta,\eta}$ and uniformly bounded on $\mathcal{R}_{\delta,\eta}$. Choose $r \in (0, \delta)$. Then $|s-1| \leq r$ implies $\mathrm{Re}(s) \geq 1 - r > 1 - \delta$, so

$$H(s, y) \ll_{\eta,\delta} 1 \qquad (|s-1| \leq r,\ |y| \leq \eta).$$

The function $q(s) = (s-1)\zeta(s)$ is holomorphic at $s = 1$ after setting $q(1) = 1$. Since $q(1) = 1 \neq 0$, we shrink r, preserving $r \in (0, \delta)$, so that q has no zeros on $|s-1| \leq r$. Then a holomorphic branch of $\log q(s)$ exists on $|s-1| < r$, normalized by $\log q(1) = 0$. Since this branch is holomorphic on a neighborhood of the closed disk $|s-1| \leq r$, it is bounded there. Therefore

$$q(s)^{2+2y} = \exp\big((2 + 2y) \log q(s)\big)$$

is uniformly bounded for $|s-1| \leq r$ and $|y| \leq \eta$. Combining this local bound with the bound for $H(s, y)$ gives $|G(s, y)| \leq C_{\eta,\delta}$.

Finally, on the punctured slit neighborhood,

$$\zeta(s)^{2+2y} = q(s)^{2+2y}(s-1)^{-2-2y},$$

with the branch of $(s-1)^{-2-2y}$ inherited from the local Hankel contour. Combining this identity with $F(s, y) = \zeta(s)^{2+2y} H(s, y)$ proves the asserted factorization. □

Lemma 5.2 (First-order approximation of the local factor)**.** Fix $0 < \eta < 1/2$ and $0 < \delta < 1/2$, and let $r \in (0, \delta)$ be chosen as in Lemma 5.1. Then

$$G(1, y) = H(1, y)$$

uniformly for $|y| \le \eta$, and

$$G(1, 0) = 1.$$

Moreover,

$$G(s, y) = G(1, y) + O_{\eta,\delta}(|s - 1|)$$

uniformly for $|s - 1| \le r/2$ and $|y| \le \eta$.

Proof. Since $q(1) = 1$ and the branch is normalized by $\log q(1) = 0$, we have

$$G(1, y) = q(1)^{2+2y} H(1, y) = H(1, y).$$

By Lemma 3.5, $H(1, 0) = 1$. Therefore $G(1, 0) = H(1, 0) = 1$.

For the first-order estimate, use Lemma 5.1. Since G is holomorphic in s on $|s - 1| < r$ and uniformly bounded for $|s - 1| \le r$, $|y| \le \eta$, Cauchy's estimate in the s-variable gives

$$\frac{\partial G}{\partial s}(s, y) \ll_{\eta,\delta} 1$$

uniformly for $|s - 1| \le r/2$ and $|y| \le \eta$. Therefore

$$G(s, y) - G(1, y) = \int_1^s \frac{\partial G}{\partial s}(w, y)\, dw = O_{\eta,\delta}(|s - 1|),$$

uniformly in the stated range. □

5.2 Evaluation of the local Hankel contribution

Proposition 5.3 (Local Hankel contribution)**.** With η fixed as in Section 4, put

$$T = \exp(\sqrt{\log x}), \qquad \rho := \frac{1}{\log x}.$$

For all sufficiently large x, uniformly for $|y| \le \eta$, we have

$$I_H = xH(1, y)\frac{(\log x)^{1+2y}}{\Gamma(2 + 2y)} + O_\eta\left(x(\log x)^{2\eta}\right).$$

Proof. Put $\alpha = 2+2y$. We apply Lemmas 5.1 and 5.2 with the fixed strip depth δ' from Section 4.3; since δ' is fixed, the constants below depend only on η. Let r be the local radius supplied by these lemmas. With $T = \exp(\sqrt{\log x})$, we have

$$\lambda_T = \frac{c_0}{2\log(T + 2)} \asymp \frac{1}{\sqrt{\log x}}.$$

Thus $\rho = 1/\log x < \lambda_T$ and $\rho < r/2$ for all sufficiently large x.

By the definition of I_H in (4.26), together with the local factorization recorded in Lemma 5.1,

$$\begin{aligned} I_H &:= \frac{1}{2\pi i}\int_{\mathcal{H}_{\mathrm{loc}}} F(s,y)\frac{x^s}{s}\,ds \\ &= \frac{1}{2\pi i}\int_{\mathcal{H}_{\mathrm{loc}}} G(s,y)(s-1)^{-\alpha}\frac{x^s}{s}\,ds \\ &= \frac{x}{2\pi i}\int_{\mathcal{H}_z} \frac{G(1+z,y)}{1+z} z^{-\alpha}e^{z\log x}\,dz, \end{aligned}$$

where the last equality follows from the change of variables $z = s-1$. Here $\mathcal{H}_z$ is the image of $\mathcal{H}_{\mathrm{loc}}$ under this change of variables. Thus $\mathcal{H}_z$ is the truncated Hankel contour whose arms run along the negative real axis from $-\lambda_T$ to $-\rho$, with the circle $|z| = \rho$ oriented counterclockwise through the right side of the origin.

For z on $\mathcal{H}_z$, Lemma 5.2 gives

$$G(1+z,y) = G(1,y) + O_\eta(|z|).$$

Also, since $|z| \le \lambda_T < 1/2$ for large x,

$$\frac{1}{1+z} = 1 + O(|z|).$$

Using the uniform boundedness of $G(1,y)$, we obtain

$$\frac{G(1+z,y)}{1+z} = G(1,y) + E(z,y), \qquad E(z,y) = O_\eta(|z|), \tag{5.4}$$

uniformly for $|y| \le \eta$ and $z \in \mathcal{H}_z$.

Accordingly, write

$$I_H = M_H + R_H,$$

where

$$M_H := \frac{xG(1,y)}{2\pi i}\int_{\mathcal{H}_z} z^{-\alpha}e^{z\log x}\,dz$$

and

$$R_H := \frac{x}{2\pi i}\int_{\mathcal{H}_z} E(z,y)z^{-\alpha}e^{z\log x}\,dz.$$

We first evaluate M_H. Put

$$U := \lambda_T \log x.$$

Under the change of variables $w = z\log x$, the contour $\mathcal{H}_z$ is mapped to a truncated Hankel contour $\mathcal{H}_U$ in the w-plane, whose arms run from $-U$ to -1 and whose circular part is $|w| = 1$. Then

$$M_H = xG(1,y)(\log x)^{\alpha-1}\frac{1}{2\pi i}\int_{\mathcal{H}_U} e^w w^{-\alpha}\,dw.$$

Let $\mathcal{H}_\infty$ be the corresponding infinite Hankel contour. By the standard Hankel representation of $1/\Gamma(\alpha)$ [],

$$\frac{1}{2\pi i}\int_{\mathcal{H}_\infty} e^w w^{-\alpha}\,dw = \frac{1}{\Gamma(\alpha)}.$$

The difference between $\mathcal{H}_\infty$ and $\mathcal{H}_U$ consists of the upper and lower tails over $(-\infty, -U]$. On these tails write $w = -u \pm i0$, with $u \geq U$. Then $|e^w| = e^{-u}$. Also, since $|y| \leq \eta$ and $\alpha = 2 + 2y$, the branch factors $e^{\pm i\pi\alpha}$ are bounded by a constant depending only on η, and hence

$$|w^{-\alpha}| \ll_\eta u^{-\operatorname{Re}(\alpha)}.$$

After taking absolute values, the two tails and the factor $1/(2\pi i)$ are absorbed into the O_η-constant, giving

$$\left| \frac{1}{2\pi i} \int_{\mathcal{H}_U} e^w w^{-\alpha}\, dw - \frac{1}{\Gamma(\alpha)} \right| \ll_\eta \int_U^\infty e^{-u} u^{-\operatorname{Re}(\alpha)}\, du.$$

Since

$$\operatorname{Re}(\alpha) = 2 + 2\operatorname{Re}(y) \geq 2 - 2\eta > 1$$

and $U \geq 1$ for sufficiently large x, we have

$$\int_U^\infty e^{-u} u^{-\operatorname{Re}(\alpha)}\, du \leq \int_U^\infty e^{-u}\, du = e^{-U}.$$

Therefore

$$\frac{1}{2\pi i} \int_{\mathcal{H}_U} e^w w^{-\alpha}\, dw = \frac{1}{\Gamma(\alpha)} + O_\eta(e^{-U}).$$

It follows that

$$M_H = xG(1, y) \frac{(\log x)^{\alpha - 1}}{\Gamma(\alpha)} + O_\eta\left(x(\log x)^{\operatorname{Re}(\alpha) - 1} e^{-U} \right).$$

Since $U = \lambda_T \log x \asymp \sqrt{\log x}$, there is a constant $c > 0$ such that $e^{-U} \ll e^{-c\sqrt{\log x}}$. Also $\operatorname{Re}(\alpha) - 1 \leq 1 + 2\eta$. Hence the tail contribution is

$$\ll_\eta x(\log x)^{1+2\eta} e^{-c\sqrt{\log x}} = O_\eta\left(x(\log x)^{2\eta} \right),$$

because $(\log x) e^{-c\sqrt{\log x}} = O(1)$.

It remains to estimate R_H. From (5.4),

$$|R_H| \ll_\eta x \int_{\mathcal{H}_z} |z|\, |z^{-\alpha}|\, e^{\operatorname{Re}(z) \log x}\, |dz|.$$

On the Hankel contour, $|\operatorname{Arg} z| \leq \pi$ and $|\Im \alpha| \leq 2\eta$, so

$$|z^{-\alpha}| \ll_\eta |z|^{-\operatorname{Re}(\alpha)}.$$

Thus

$$|R_H| \ll_\eta x \int_{\mathcal{H}_z} |z|^{1 - \operatorname{Re}(\alpha)} e^{\operatorname{Re}(z) \log x}\, |dz|.$$

Changing variables again by $w = z \log x$ gives

$$|R_H| \ll_\eta x(\log x)^{\operatorname{Re}(\alpha) - 2} \int_{\mathcal{H}_U} |w|^{1 - \operatorname{Re}(\alpha)} e^{\operatorname{Re} w}\, |dw|.$$

The last integral is $O_\eta(1)$: this is immediate on $|w| = 1$, and on the arms it is bounded by

$$\int_1^U u^{1 - \operatorname{Re}(\alpha)} e^{-u}\, du \leq \int_1^\infty u^{1 - \operatorname{Re}(\alpha)} e^{-u}\, du \ll_\eta 1.$$

Consequently,

$$R_H = O_\eta \left(x(\log x)^{\mathrm{Re}(\alpha)-2} \right).$$

Since $\alpha = 2 + 2y$, we have

$$\mathrm{Re}(\alpha) - 2 = 2\,\mathrm{Re}(y) \le 2\eta,$$

and hence

$$R_H = O_\eta(x(\log x)^{2\eta}).$$

Combining the estimates for M_H and R_H, and using Lemma 5.2 to replace $G(1, y)$ by $H(1, y)$, we obtain

$$I_H = xH(1, y)\frac{(\log x)^{1+2y}}{\Gamma(2 + 2y)} + O_\eta(x(\log x)^{2\eta}),$$

uniformly for $|y| \le \eta$. □

Proposition 5.4 (Generating-function asymptotic)**.** With the parameters chosen as in Sections 4 and 5, and with the fixed contour parameters suppressed in the implied constant, uniformly for $|y| \le \eta$, as $x \to \infty$,

$$A(x, y) = xH(1, y)\frac{(\log x)^{1+2y}}{\Gamma(2 + 2y)} + O_\eta \left(x(\log x)^{2\eta} \right).$$

Proof. For non-integral x, the contour decomposition (4.24) gives

$$A(x, y) = I_H + I_{\mathrm{top}} + I_{\mathrm{bot}} + I_{\mathrm{left}} + E_{\mathrm{Perron}}(x, T, y).$$

By Proposition 4.10, for some $c_1 > 0$,

$$E_{\mathrm{Perron}}(x, T, y) + |I_{\mathrm{top}}| + |I_{\mathrm{bot}}| + |I_{\mathrm{left}}| \ll_\eta x \exp(-c_1\sqrt{\log x}),$$

uniformly for $|y| \le \eta$. Proposition 5.3 gives

$$I_H = xH(1, y)\frac{(\log x)^{1+2y}}{\Gamma(2 + 2y)} + O_\eta \left(x(\log x)^{2\eta} \right).$$

The exponentially small nonlocal and Perron error is absorbed into $O_\eta(x(\log x)^{2\eta})$. Therefore the asserted asymptotic holds for non-integral x.

For integral x, the replacement by $x + 1/2$ described in Section 4.1 gives the same estimate, since $A(x, y) = A(x + 1/2, y)$ and replacing $x + 1/2$ by x changes the displayed main term by an amount absorbed into the same error term. □

6 Extraction of $T_j(x)$ and proof of the main theorem

Proof of Theorem 1.3. By Proposition 5.4, uniformly for $|y| \le \eta$,

$$A(x, y) = xH(1, y)\frac{(\log x)^{1+2y}}{\Gamma(2 + 2y)} + O_\eta \left(x(\log x)^{2\eta} \right).$$

Define

$$K(y) := \frac{H(1, y)}{\Gamma(2 + 2y)}.$$

Then K is holomorphic in a neighborhood of 0, and

$$A(x,y) = x\log x\, K(y)e^{2y\log\log x} + O_\eta\left(x(\log x)^{2\eta}\right).$$

The generating relation

$$A(x,y) = \sum_{j\ge 0} T_j(x)y^j$$

gives the coefficient identity below. Here $[y^j]$ denotes the coefficient of y^j in the power series expansion:

$$T_j(x) = [y^j]A(x,y).$$

Write

$$K(y) = \sum_{r\ge 0} k_r y^r, \qquad k_r = \frac{K^{(r)}(0)}{r!},$$

and

$$e^{2y\log\log x} = \sum_{m\ge 0} \frac{(2\log\log x)^m}{m!}y^m.$$

For fixed j,

$$[y^j]\left(K(y)e^{2y\log\log x}\right) = \sum_{r=0}^{j} k_r \frac{(2\log\log x)^{j-r}}{(j-r)!}.$$

Therefore

$$[y^j]\left(K(y)e^{2y\log\log x}\right) = k_0\frac{(2\log\log x)^j}{j!} + k_1\frac{(2\log\log x)^{j-1}}{(j-1)!} + O_j\left((\log\log x)^{j-2}\right).$$

It remains to compute k_0 and k_1. By Lemma 3.5, $H(1,0)=1$. Also $\Gamma(2)=1$. Hence

$$k_0 = K(0) = 1.$$

Next,

$$k_1 = K'(0) = \partial_y H(1,0) + H(1,0)\left.\frac{d}{dy}\frac{1}{\Gamma(2+2y)}\right|_{y=0}.$$

Here $\psi = \Gamma'/\Gamma$ denotes the digamma function. Since

$$\left.\frac{d}{dy}\frac{1}{\Gamma(2+2y)}\right|_{y=0} = -2\psi(2) = -2(1-\gamma),$$

we get

$$k_1 = \partial_y H(1,0) - 2(1-\gamma).$$

We also need to extract coefficients from the error term. Choose a fixed radius ρ_y with $0<\rho_y<\eta$. Since the error in Proposition 5.4 is uniform for $|y|\le\eta$, Cauchy's coefficient estimate on $|y|=\rho_y$ gives, for each fixed j,

$$[y^j]\,O_\eta\left(x(\log x)^{2\eta}\right) = O_{\eta,j,\rho_y}\left(x(\log x)^{2\eta}\right).$$

Because $0<\eta<1/2$, for fixed $j\ge 2$,

$$x(\log x)^{2\eta} = O_j\left(x\log x(\log\log x)^{j-2}\right).$$

Thus this error is absorbed into the final error term.

Combining the preceding estimates gives, for each fixed $j \geq 2$,

$$\begin{aligned} T_j(x) = {} & \frac{2^j}{j!} x \log x (\log\log x)^j \\ & + \frac{2^{j-1}}{(j-1)!} \left(\partial_y H(1,0) - 2(1-\gamma)\right) x \log x (\log\log x)^{j-1} \\ & + O_j\left(x \log x (\log\log x)^{j-2}\right). \end{aligned}$$

In particular,

$$T_j(x) \sim \frac{2^j}{j!} x \log x (\log\log x)^j.$$

□

7 Toward $D(x)$: outlook

The identity (1.6) expresses $D(x)$ exactly in terms of the prime-power contribution and the alternating finite sum of the quantities $T_j(x)$. Theorem 1.3, however, is a fixed-j asymptotic statement. Since the range of relevant j grows with x, these fixed-j asymptotics cannot be inserted term by term into (1.6) to obtain an asymptotic formula for $D(x)$.

Equivalently, if

$$A(x,y) = \sum_{j \geq 0} T_j(x) y^j,$$

then the alternating part of (1.6) is determined by values of $A(x,y)$ and its y-derivative at $y = -1$. The analytic work in this paper is carried out for y in a small neighborhood of 0. Extending the method toward $D(x)$ would therefore require estimates uniform in j, or a direct analysis of the generating function near $y = -1$.

References

[1] Tom M. Apostol. *Introduction to Analytic Number Theory*. Undergraduate Texts in Mathematics. Springer-Verlag, New York, 1976.

[2] P. G. L. Dirichlet. Über die Bestimmung der mittleren Werthe in der Zahlentheorie. *Abhandlungen der Königlich Preussischen Akademie der Wissenschaften zu Berlin*, pages 69–83, 1849.

[3] G. H. Hardy. On dirichlet's divisor problem. *Proceedings of the London Mathematical Society*, 15:1–25, 1916.

[4] M. N. Huxley. Exponential sums and lattice points iii. *Proceedings of the London Mathematical Society*, 87(3):591–609, 2003.

[5] Georges Voronoï. Sur un problème du calcul des fonctions asymptotiques. *Journal für die reine und angewandte Mathematik*, 126:241–282, 1903.